Geophys. J. Int. (2007) **171,** 478–494 doi: 10.1111/j.1365-246X.2007.03539.x

Gravimetric determination of an intrusive complex under the Island of Faial (Azores): some methodological improvements

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Accepted 2007 June 26. Received 2007 June 26; in original form 2006 September 6

SUMMARY

We present some improvements of a gravity inversion method to determine the geometry of the anomalous bodies for priori density contrasts. The 3-D method is based on an exploratory process applied, not for the global model, but for the steps of a growth approach. The (positive and/or negative) anomalous structure is described by successive aggregation of cells, while its corresponding gravity field remains nearly proportional to the observed one. Moreover, a simple (e.g. linear) regional trend can be simultaneously adjusted. The corresponding program is applied to new gravity data on the volcanic island of Faial (Azores archipelago). The inversion approach shows a subsurface anomalous structure for the island, the main feature being an elongated high-density body. The body is interpreted as a compact sheeted dyke swarm, emplaced along Faial-Pico Fracture Zone, a leaky transform structure that forms the current boundary between Eurasian and African plates in the Azores area. The new results in this paper are (1) a Bouguer gravity anomaly map, (2) several improvements in the inversion process (robust process, optimal balance fitness/model magnitude), (3) a new gravimetric method for estimating the mean terrain density, (4) a 3-D model for subsurface mass anomalies in Faial and (5) some interpretative conclusions about a main intrusive complex detected under the island as a wall-like structure extending from a depth of 0.5 to 6 km b.s.l., with a N100°E trend and corresponding to an early fissural volcanic episode controlled by the regional tectonics.

Key words: Faial (Azores), gravity anomalies, inverse problem, volcanic structure.

GJI Volcanology, geothermics, fluids and rocks

1 INTRODUCTION

According to Walker (1999), rift zones and underlying dyke swarms occur in most volcanoes and contain the paths taken by magmas moving through the crust. Most of these intrusive complexes are positioned and oriented by tectonic structures or may be propagated laterally from volcanic centres along rift zones, following neutral buoyancy levels.

Sheeted dyke complexes are detected in oceanic-spreading settings and in the core regions of major volcanoes. They may contain thousands of very narrow dykes or other sheet-like intrusions as swarms that increase abruptly in intensity at their edges. Dyke complexes are perceived to grow incrementally by the addition of dykes along their margins, and are self-sustaining (Walker 1999). Additionally, incoming magma batches are guided by the many planes of weakness in the complex. Closer to the surface, swarms of sheet intrusions (dykes and sills) represent conduits for magma transport from the reservoirs to the surface or to shallow intrusions, the setting of this latter type of intrusions being well documented in active volcanoes, such as Kilauea or Piton de la Fournaise (Malengreau *et al.* 1999). Walker (1999) also suggests that intrusion of dykes may cause bending or initiation of a rift zone.

Large swarms of intrusive bodies can be gravimetrically determined. Accordingly to Rymer & Brown (1986), positive anomalies characterize mainly basaltic volcanoes, and are caused by a relatively dense intrusive complex/magma body, which contrasts with its surroundings either because that intrusive body is more mafic than average or, more likely, because near surface, previously erupted materials are uncompacted, with a higher degree of vesiculation. Walker (1992) for Hawaiian volcanoes considers an intrusive complex with density 2800 kg m^{-3} juxtaposed against highly vesicular lava flows having a much lower density of about 2000 kg m^{-3}. Ryan (1987 in Malengreau *et al.* 1999) proposed that complexes of hypovolcanic intrusions and cumulates develop upwards during growth of a volcano. As a result, the centre of a mature oceanic shield volcano is expected to be characterized by a column-like body of high-density rock that can be detected through gravimetric observation and modelled by some inversion approach. Apart from the general ambiguity problem of the gravity inversion, some problems arise from the accessibility and sharp topographic

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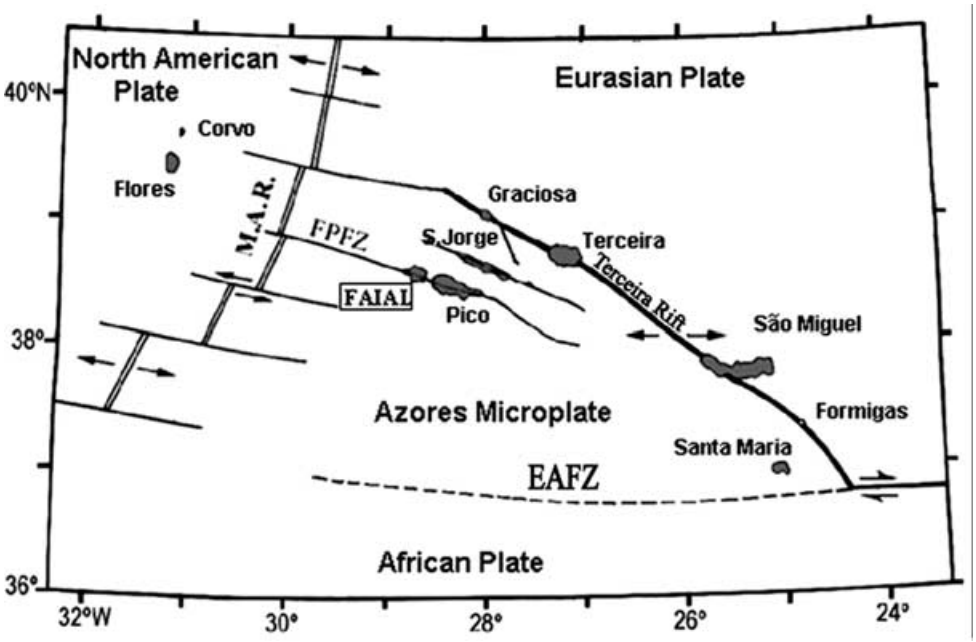

Figure 1. General geotectonic framework of Azores archipelago, at the triple junction of Eurasian, North American and African plates. Location of Faial Island is highlighted. M.A.R.: Mid-Atlantic Ridge; FPFZ: Faial-Pico Fracture Zone; EAFZ: East Azores Fracture Zone. Simplified from: Montesinos *et al.* (2003).

effects in volcanic areas mainly formed by very high bulk density material.

The Azores archipelago is located in the North Atlantic Ocean, between latitudes 37° and 40°N and longitudes 25° and 31°W, at the triple junction of Eurasian, North American and African lithospheric plates (Fig. 1). It consists of nine islands and several islets of volcanic nature that emerge from an anomalously shallow and rough topographic zone, the so-called 'Azores Plateau' (Needham & Francheteau 1974), which is roughly triangular-shaped and is defined by the 2000 bathymetric contour line. This area broadly coincides with the Azores 'microplate'or 'block', bordered by the Mid-Atlantic Ridge (M.A.R.), the Terceira Rift and the East Azores Fracture Zone (EAFZ, Fig. 1), and it is characterized by recent and active volcanoes and high seismicity (Searle 1980; Luis *et al.* 1994; Lourenço *et al.* 1998; Luis *et al.* 1998). The islands are aligned along major tectonic lineaments with a general WNW–ESE trend, with the M.A.R. placed between Faial and Flores islands (Fig. 1).

Faial Island is located about 120 km east of the M.A.R., on the seismically active Faial-Pico Fracture Zone (FPFZ, Fig. 1). This structure is a 350 km long leaky transform that extends from the M.A.R. with an ESE trend, and is considered by some (e.g. Luis *et al.* 1994) as the third arm of the Azores triple junction and, therefore, the present boundary between the Eurasian and African plates.

The following sections present a process for determining a well-defined intrusive complex beneath the volcanic island of Faial. First, we present information about the geology and tectonics of the island. Secondly, we describe the gravity survey and the resulting Bouguer anomaly map. Then, we present the inversion method applied to the gravity anomaly. This method is based on an exploratory solution of the general non-linear three-dimensional (3-D) problem using inexact data. A simulation test is added for better understanding of the process. The application of the method to the anomaly of Faial produces a 3-D model of anomalous masses. This model shows the presence of an interesting high-density body located on the central-eastern part of Faial Island. The final sections discuss the resulting density structure model and its implications for an intrusive complex beneath the island.

2 GEOLOGY AND VOLCANOLOGY OF FAIAL ISLAND

The geology of Faial island (21 km length and 173 km^2) is in general terms dominated by a central volcano with caldera ('Caldeira Volcano') and by a western peninsula ('Capelo' Peninsula) built by 20 scoria cones and related basaltic lava flows (Fig. 2). Near the town of Horta, about one dozen scoria cones are emplaced along NW–SE trending fractures and are all covered with pumice deposits from the central volcano caldera. The eastern part of the island is characterized by the Pedro Miguel Graben, which has a N115°E average trend and width of about 7 km. In this eastern area, near Espalamaca, are the oldest rocks of Faial, dated 730 000 yr before present (BP) by Feraud *et al.* (1980). They are related to an old shield volcano—the 'Ribeirinha' Volcano—centred east of 'Caldeira' and are covered by its younger trachytic products (Figs 2 and 3).

The 'Caldeira Volcano' has a maximum altitude of 1043 m, average base diameter of 14 km, area of 133 km^2, volume of about 48 km^3 (Nunes *et al.* 2004) and is built mostly by lava flows of basaltic to benmoreitic composition. Its late eruptive history during the past 10 000 yr (Madeira 1998; Pacheco 2001) is characterized by explosive trachytic eruptions, accompanied by voluminous pumice fall deposits, ignimbrites and lahars (Serralheiro *et al.* 1989). During those late explosive eruptions, a caldera 2 km wide and about 400 m deep was formed (Fig. 2).

Basaltic volcanism dominates on the Capelo peninsula, with about twenty Holocene eruptions, the last being the 1957–1958 Capelinhos surtseyan eruption, located on its westernmost end. That eruption added 1.5 km^2 of new land to the island, and gave rise to 1.5 m maximum subsidence during the May 1958 seismic crisis (Machado 1958; Tazieff 1959; Machado *et al.* 1962), as well as the opening of several tension cracks.

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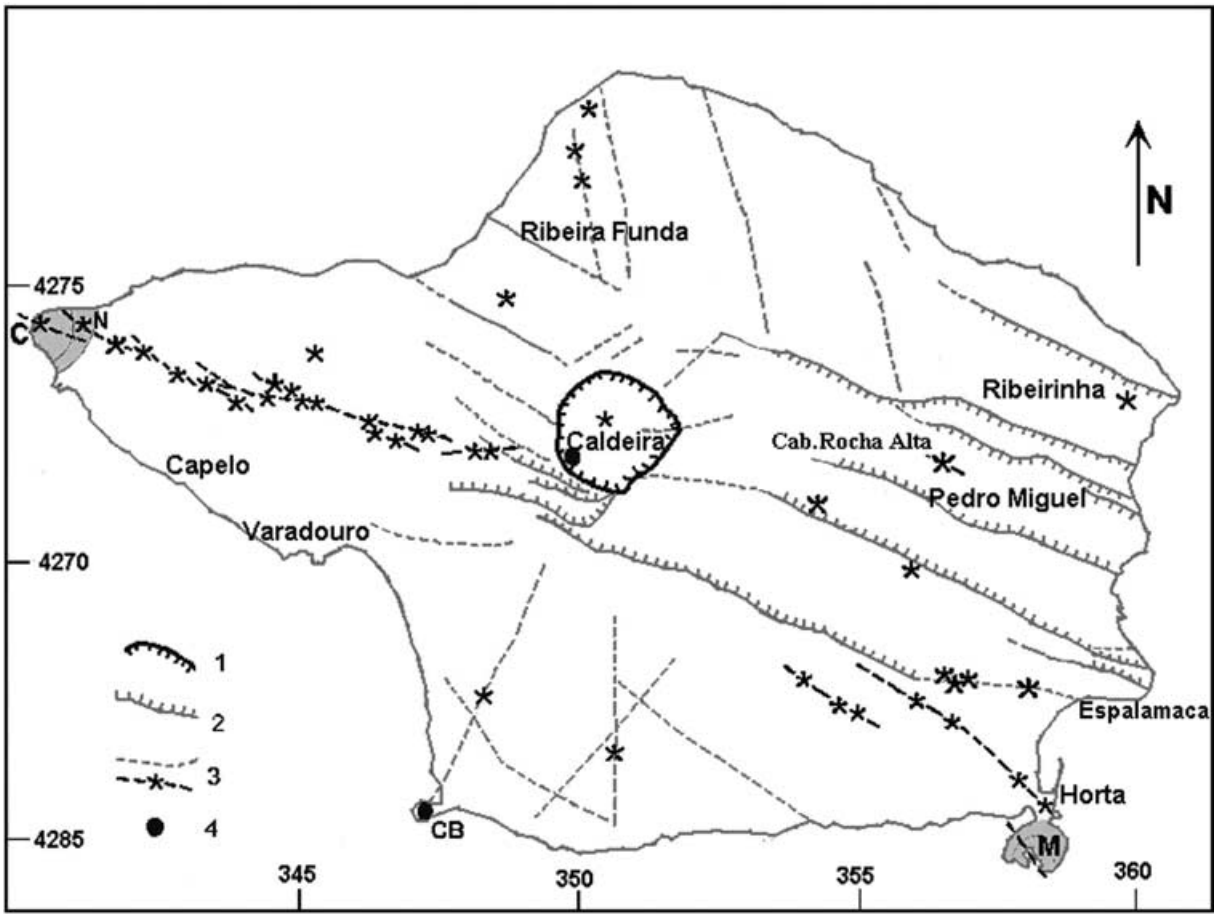

Figure 2. Main volcanic and tectonic features of Faial island: (1) caldera rim; (2) Pedro Miguel Graben faults; (3) main volcanotectonic lineaments, including scoria cone vents (∗); (4) Castelo Branco (CB) and Altar (at Caldeira) trachytic domes; in grey- tuff cones of Monte da Guia (M), Costado da Nau (N) and Capelinhos (C). Modified after Serralheiro *et al.* (1989) and Madeira (1998).

As mentioned, the 'Pedro Miguel Graben' is the most prominent feature in the eastern part of Faial Island: its fault scarps face SW and NE, with maximum height of about 170 m. The lowest block of that structure is located near Pedro Miguel village (Fig. 2). Close to the central volcano summit the graben is mostly covered/filled by pumice and other Holocene pyroclastic deposits, but it can be traced again west of the 'Caldeira', namely at Ribeira Funda fault, even less clear on the topography (Fig. 3). Pedro Miguel fault scarps started developing during the last 73 000 (or 40 000) yr, then channelled the pyroclastic flow deposits (e.g. lahars and ignimbrites) extruded during the 'Caldeira' episodes (Madeira 1998). According to Tazieff (1959), Pedro Miguel Graben is merely a pure tectonic structure that influenced the volcanism on Faial Island. According to MacDonald (1972) the graben is the result (1) of the removal of magma from an underlying magma chamber, (2) the result of stretching of the surface of the volcano or (3) the result of stretching of the entire underlying crust, due to the spreading associated with the Atlantic Ocean.

3 GRAVITY DATA AND ANOMALY

In 2000, a gravity survey was conducted on the island of Faial on 253 stations (Fig. 4) with a mean spacing of 800 m. A total of 274 observations were carried out using a LaCoste&Romberg gravimeter (G665) with electronic output. Observations were corrected for tidal variations. Then we determined an instrument drift by fitting traverses with a mean residual value about 0.02×10^{-5} m s^{-2} for repeated stations. Gravity values were referenced to the absolute gravity value $g = 980\,129\,217\ (\pm 0.007) \times 10^{-5}$ m s^{-2} determined in 1997 with the GILAg-5 of the Finnish Geodetic Institute (Bastos *et al.* 1999) on the seismic pillar of the Meteorological Observatory of Horta (Faial) (main gravity base station for the island, Coelho 1968).

A contemporaneous GPS differential survey was applied to determine station coordinates, mainly elevations. Observations were carried out with an Ashtech Z-Surveyor equipment and, for the final fit, we employed geodetic values for eight bench-marks as a reference network on the island. The estimated accuracy of the resulting elevation values is about 0.05 m.

As part of gravity processing, we used a digital terrain model (DTM) for the island formed by 69 474 points arranged on a grid with 50 m resolution. In addition, we consider the bathymetry for the neighbouring areas (within a distance of 80 km) coming from the data files of global topography in SRTM30 format distributed by the USGS EROS data centre, and corresponding to a grid with 30 s resolution (http://topex.ucsd.edu/WWW_html/srtm30_plus.html; Smith & Sandwell 1997). This terrain model was used for the relief features in Fig. 3. Taking into account the rough resolution of the DTM, some very local residual effects can be suspected after the terrain correction.

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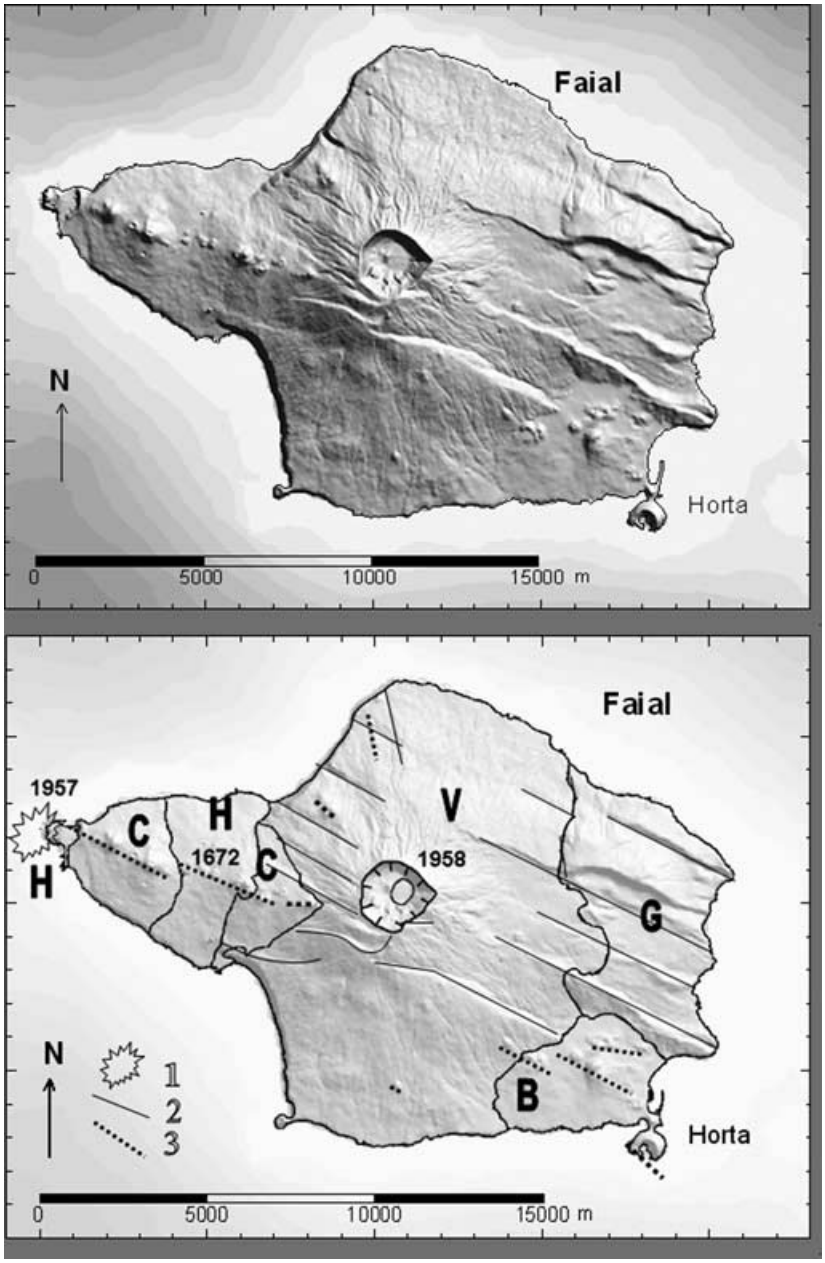

Figure 3. DEM for Faial Island (up) and superimposed main volcanotectonic structures of the island (down). (1) Capelinhos Volcano, (2) faults and (3) main volcanotectonic lineaments. General volcanostratigraphy: H: Historical eruptions (with year of occurrence), C: Capelo Volcanic Complex, V: Caldeira Volcano, B: Horta Basaltic Zone, G: Ribeirinha Volcano and Pedro Miguel Graben (adapted from Nunes, 2004).

Using the gravity, position and terrain data we calculated 'station completed' Bouguer anomaly values (including terrain correction). We used a density of 2341 kg m^{-3} for the Bouguer and terrain corrections, a value which is typically found in density determinations of rocks of the Azores islands (see Motta & Nunes 2003; Montesinos *et al.* 2003; Nunes *et al.* 2006) and, mainly, according to a gravimetrical estimation ($\pm$21 kg m^{-3}) explained later in the paper. Fig. 4 shows the resulting Bouguer gravity values, as well as the Bouguer anomaly values of 75 gravity stations from the neighbouring island of Pico (Camacho *et al.* 1999; França 2000) to improve the regional trend fit of the Faial data in subsequent analysis. The resulting anomaly values range between 87×10^{-5} and 120×10^{-5} m s^{-2}, with a strong low on the western edge (Capelinhos Volcano) and an elongated maximum over the northeastern sector of Faial island (Pedro Miguel graben), the latter with a N100°E general trend and southern border roughly along the Espalamaca fault (Fig. 2). In the next sections, we construct a model of anomalous masses responsible for this observed positive gravity anomaly. As obtained below by applying an analysis of autocorrelation in the inversion process, the estimated noise level for the gravity anomaly values is 0.6×10^{-5} m s^{-2}. This noise value corresponds to the data quality, but mainly to short-wavelength gravity anomalies and short-wavelength terrain effects.

4 INVERSION METHOD

The inverse gravimetric problem, namely the determination of a subsurface mass density distribution corresponding to an observed

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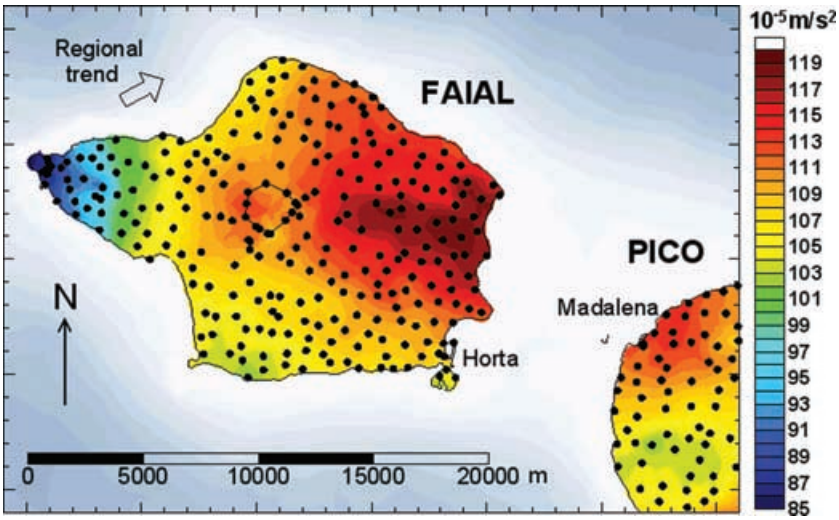

Figure 4. Bouguer anomaly in Faial (terrain density 2390 kg m^{-3}) supplemented with data from Pico.

gravity anomaly, has an intrinsic non-uniqueness in its solution (e.g. Al-Chalabi 1971). Moreover, data must be considered as insufficient and inaccurate. Nevertheless, particular solutions can be obtained by including additional constraints about the model parameters (subsurface structure) and about the data parameters (statistical properties of the inexact data, e.g. Gaussian distribution). The inversion methods that define the geometrical properties of anomalous bodies with prescribed density contrast (e.g. Pedersen 1979; Barbosa *et al.* 1997) correspond to a non-linear methodology and offer interesting results, limited to the validity of the hypothesis used. Unfortunately, linearized techniques depend strongly on the accuracy of initial estimates of the model parameters (Rothman 1985). For the fully non-linear treatment, the methods of exploration of the space model often give the best option (Tarantola 1988). This exploration process can be conducted randomly (Silva & Hohmann 1983) or systematically.

A general tool for describing the irregular geometry of the anomalous mass structure is obtained by means of aggregation of small parallelepiped cells filled with anomalous mass as a mosaic. This procedure can describe general 3-D models (including some relief details and spherical effects) but gives rise to a very large number of degrees of freedom for the model. In fact, it would suppose one parameter for each cell, in a model composed of several thousand cells. Therefore, a general exploratory inversion approach, simultaneously for every cell, would be ineffective. An interesting idea was proposed by René (1986). He applied (in a more restrictive context) an exploratory method, not to try every possible density distribution for the complete model (too hard a task), but to try only the different options for each step of a process for building the model in a growth approach from initial 'seeds'. In these circumstances, for each step of the model growth (by filling or adding a new cell), the number of degrees of freedom is drastically reduced, so the exploratory process becomes very efficient.

René (1986) put forward a 2-D expansion approach by using a 'maternal' structure formed by square 'seeds' that grow by incorporating only contiguous elements. The René method does not require additional constraints or hypotheses but uses only positive (or only negative) density contrasts, and then models corresponding to both positive and negative anomalies are not accepted. On the basis of this pioneering work, Camacho *et al.* (2000, 2002) presented a gravity inversion approach with the following features: (1) considered a 3-D context, (2) non-gridded non-planar inaccurate data were accepted, (3) a 'maternal' or 'seed' structure was not required, (4) a simple regional trend was simultaneously determined, (5) matter expansion can appear anywhere (not only for contiguous elements), (6) not requiring 'contiguous' expansion enabled us to consider non-regular subsoil partition (e.g. with deeper blocks bigger than shallow blocks), (7) if previous qualified models (density distribution plus corresponding covariance matrix) exist, they can be incorporated and above all (8) positive and negative density contrasts are simultaneously accepted. The last two improvements were achieved by means of an additional condition to the model. Tiede *et al.* (2005) and Camacho *et al.* (2001) present some study cases corresponding to volcanic structures.

This paper presents a new version of the gravity inversion method with several improvements: (1) a robust treatment is adopted for the gravity data, that considers outliers, (2) an *a priori* self-variable density contrast is accepted for smooth models, (3) the terrain density for topographic correction is included to be simultaneously estimated and (4) a criterion for optimal choice of the balance between fitness and smoothness of the model is proposed (equivalent to a criterion to estimate the gravity data noise level). We will now detail the method.

Let us consider n gravity stations P_i (x_i, y_i, z_i), $i = 1, \ldots, n$, not necessarily gridded, located on rugged topography and with observed Bouguer anomaly values Δg_i (corrected for topographic or terrain effects according to a terrain density value ρ_T). We suppose a mostly Gaussian distribution for the observation uncertainties given by a covariance matrix $\boldsymbol{Q_D}$ (as deduced from data analysis). Also we accept a few erroneous gravity values. Our goal is to construct a 3-D model of the subsurface bodies, for prescribed density contrasts that can give rise to the observed anomaly. To that end, the subsurface volume close to the survey area is broken down into a global discrete 3-D partition of m prismatic elements. The desired solution will be described as an aggregation of some cells filled with prescribed density contrast (meanwhile other cells will remain 'empty' of anomalous mass). The resolution degree of the model (number of elemental cells describing the geometry of anomalous bodies) is related to the size of the cells. This depends on the gravity data distribution. In fact, for an area covered with close stations the size of

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the cells could be smaller than for an area with far stations. Finally, the number of cells and the resolution of the model are limited by our computing capacity.

As usual for these non-linear methods, the few values of density contrasts that are going to be used to constitute the anomalous model (by filling some cells) are previously prescribed. For the *j*th cell we decide an *a priori* positive contrast $\Delta\rho_j^+$ (if excess of mass is finally fitted there) and an *a priori* negative contrast $\Delta\rho_j^-$ (if defect of mass is finally fitted). In Section 5.1, we further detail how the starting values for $\Delta\rho_j^+$ and $\Delta\rho_j^-$ have been chosen. The possibility of non-anomalous mass ('empty' cell) is the third option.

The gravity attraction, A_{ij}, at the *i*th station $P_i(x_i, y_i, z_i)$, due to the *j*th prism, for unit density, is given by Pick *et al.* (1973):

$$A_{ij} = -G\Big\{\big[(\mathrm{x}\ln(\mathrm{y} + (\mathrm{x}^2 + y^2 + z^2)^{1/2}) + y\ln(\mathrm{x} + (\mathrm{x}^2 + y^2 + z^2)^{1/2}) + \mathrm{z}\,\mathrm{arctg}(\mathrm{z}(\mathrm{x}^2 + y^2 + z^2)^{1/2}x^{-1}y^{-1})\big)_{u_1^j - x_i}^{u_2^j - x_i}\Big]_{v_1^j - y_i}^{v_2^j - y_i}\Big\}_{w_1^j - z_i}^{w_2^j - z_i}, \tag{1}$$

where G is the gravitational constant, the edges of the *j*th prism are parallel to the reference axes, and the limiting coordinates for its volume are: u_1^j, u_2^j for the *x* coordinates, $v^j{}_1$, v_2^j for the *y* coordinates and $w^j{}_1$, w_2^j for *z*. A 3-D mosaic with prisms can approximate the irregular volume to study. Matrix $\boldsymbol{A}$, with components A_{ij} is the design matrix of the physical configuration of the problem and contains the effect of the rugged terrain, the station distribution, the partition of subsurface volume, etc. Now the resulting model must reproduce the observed anomaly (except the observation noise):

$$\Delta g_i = \sum_{j\in J^+} A_{ij}\,\Delta\rho_j^+ + \sum_{j\in J^-} A_{ij}\,\Delta\rho_j^- + \delta g_{\mathrm{reg}} + \delta g_{\mathrm{top}} + v_i, \quad i = 1, \ldots, n, \tag{2}$$

where J^+, J^- are the sets of indexes corresponding to the cells filled with positive and negative density contrasts, δg_{reg} represents a component corresponding to a regional smooth trend, δg_{top} is an additional term for revision of the topographic correction and v_i is a residual value which represents the local noise. It is notable that we include the regional trend and the topographic correction (or, better, a corrective term to the topographic term used in the gravity reduction) as a part of the inversion approach instead of a treatment before inversion. For simplicity, we are going to adopt a linear expression for the regional trend component δg_{reg}:

$$\delta g_{\mathrm{reg}} = p_0 + p_x(x_i - x_M) + p_y(y_i - y_M), \quad i = 1, \ldots, n, \tag{3}$$

where x_M, y_M are the planar (UTM, for instance) coordinates of an arbitrary central point for the survey, and p_0, p_x, p_y are three unknown values which adjust a trend. A 1-degree polynomial surface is chosen because it simplifies the subsequent formulation and usually it is able to properly fit the very long wavelength component of the data. However, higher degree polynomial surfaces can be implemented in a similar way. Below (Section 5.5) we give some additional comments about the regional trend. On the other hand, the topographic correction term will have the following expression:

$$\delta g_{\mathrm{top}} = \delta\rho_{\mathrm{T}}\, C_i, \tag{4}$$

where $\delta\rho_{\mathrm{T}}$ will be the unknown terrain density (or, better, a correction of a terrain density previously adopted in the gravity reduction) and C_i will be a coefficient depending on the relief geometry around point $P_i(x_i, y_i, z_i)$. Below, we give more details about these two corrective terms. So on, J^+, J^-, p_0, p_x, p_y and $\delta\rho_{\mathrm{T}}$ are the main unknowns to be determined in this inversion process.

Previous information about the model structure, if known, can be optionally incorporated. To that end, we express this information as initial values ρ_j^o, $j = 1, \ldots, m$, for the prism densities and a corresponding covariance matrix $\boldsymbol{Q_M}$ corresponding to the previous model confidence (Tarantola 1988) (bold character will be used for matrices and column vectors). For a problem without previous information about the model structure, we can adopt $\rho_j^o = 0$, $j = 1, \ldots, m$, and take a model covariance matrix $\boldsymbol{Q_M}$ given by a diagonal normalizing matrix of non-null elements that are the same as the diagonal elements of $\boldsymbol{A}^T\boldsymbol{Q_D}^{-1}\boldsymbol{A}$. (The elements of this matrix can be used also to limit the whole subsurface volume in study, bounding the accepted model uncertainty). The gravity values calculated for this initial density structure at points P_i, $i = 1, \ldots, n$, are:

$$\Delta g_i^0 = \sum_{j=1}^{m} A_{ij}\,\rho_j^0 \tag{5}$$

to be removed from the observed anomaly.

If only a minimization criterion for the residuals $\boldsymbol{v}$ is used, the acceptance of positive and negative values for the prescribed density contrasts and the inclusion of the trend unknowns give an inherent non-unique solution. To solve this, an additional condition of minimization of the model variation can be adopted. Thus, the solution is obtained by a mixed condition formed by the gravity l_2-fitness and the whole anomalous mass quantity, using a parameter λ for the suitable balance:

$$\mathbf{v}^T\,\mathbf{Q}_\mathbf{D}^{-1}\,\mathbf{v} + \lambda\,\mathbf{m}^T\,\mathbf{Q}_\mathbf{M}^{-1}\,\mathbf{m} = \min, \tag{6}$$

where $\mathbf{v} = (v_{1_1}, \ldots, v_n)^{\mathrm{T}}$ (T for transpose) are the gravity residuals for the N stations, $\mathbf{m} = (\Delta\rho_{1_1}, \ldots, \Delta\rho_m)^{\mathrm{T}}$ is the anomalous density for the m cells of the subsurface volume partition $\Delta\rho_j^+$ or $\Delta\rho_j^-$ for the filled cells and zero for those not filled), λ is a positive factor, empirically fixed, for balance between model fitness and anomalous model magnitude (and complexity). $\boldsymbol{Q_D}$ is the covariance matrix (usually diagonal matrix) corresponding to the estimated (Gaussian) inaccuracies of the gravity data, and $\boldsymbol{Q_M}$ is the covariance matrix corresponding to the supposed determinability of the model parameters $\boldsymbol{m}$. The first addend of the minimization functional (6) corresponds to the fit residuals weighted with the data quality matrix. The second addend is a weighted addition of the model densities. Nevertheless, taking into account that the covariance matrix $\boldsymbol{Q_M}$ contains the inverse values of the squared prism volumes as factor, this second addend is connected with the anomalous mass or magnitude of the model.

The λ parameter governs the application of the minimization conditions regarding the balance between total anomalous mass and residual values. For low λ values a good fit is obtained, but the mass of the anomalous model may increase excessively and fictitious structures may appear, mainly trying to fit the data noise. For high λ values the adjusted model can be too slight, with too little mass, and a poor gravity fit is obtained. Below we propose a practical criterion for choosing optimal λ values and then determining the noise level in the gravity data.

The inversion process seeks to determine an anomalous model geometrically, described as an aggregation of cells filled with the *a priori* density contrasts (positive and negative), so that it verifies the minimization condition (6). As previously pointed out, we address this non-linear problem with a process that explores the large number of possible models looking for the best one. A pure explorative method would consider every (!) possible model, trying their fit to

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the minimization conditions. The highly tedious task of global exploration can be avoided by using a 'growth' process or expansion approach to construct the anomalous bodies. The bodies are formed in a nearly homothetic growth by adding cells, from former 'skeletal' structures, also adjusted, until they attain a suitable size. Then, the exploration of *all the model's possibilities* is substituted by exploration of the several *possibilities of growth* (cell by cell) for each step of the anomalous bodies expansion. This is a more reasonable goal. Thus, the prismatic cells are systematically tested, step by step, with each prescribed density, and then the best options are adopted and added to the growth approach for the anomalous bodies set-up. The minimization fit conditions are applied for each growth step, now including a scale factor f which relates the immature model to the global conditions concerning gravity fit and model magnitude. Further details are given below.

For an arbitrary $(k+1)$th step, k prisms have been previously filled with the positive or negative fixed contrast values (or changed from some initial values) and the modelled gravity values will be:

$$\Delta g_i^c = \Delta g_i^0 + \sum_{J_k^+} A_{ij}\,\Delta\rho_j^+ + \sum_{J_k^-} A_{ij}\,\Delta\rho_j^-, \tag{7}$$

where J_k^+, J_k^- are the index sets corresponding to the previously modified prisms with densities $\rho_j^o + \Delta\rho_j$. Now, we look for one new prism to modify throughout the m–k unchanged prisms. For each jth unchanged prism, $j \notin J_k^+, J_k^-$, the following equation system can be considered:

$$\Delta g_i - \left(\Delta g_i^c + A_{ij}\,\Delta\rho_j\right) f - [p_0 + p_x(x_i - x_M) + p_y(y_i - y_M)] - \delta\rho_T C_i = v_i \quad i = 1, \ldots, n, \tag{8}$$

where $\Delta\rho_j$ can take values $\Delta\rho_j^+$ and $\Delta\rho_j^-$ and $f \geq 1$ is an unknown scale factor for fitting the 'actual' model anomalies $(\Delta g_i^c + A_{ij}\,\Delta\rho_j)$ to the observed ones, Δg_i. The positive and the negative prescribed values are successively tested for the additional density contrast $\Delta\rho_j$ looking for the minimization condition (6).

Next, the unknown parameters f, p_0, p_x, p_y and $\delta\rho_{\mathrm{T}}$ are adjusted for a mix minimization criterion corresponding to this $(k+1)$th step:

$$\mathbf{v}^T \mathbf{Q}_{\mathrm{D}}^{-1}\mathbf{v} + \lambda\, f^2\, \mathbf{m}^T\, \mathbf{Q}_M^{-1}\, \mathbf{m} = \min, \tag{9}$$

where the vector $\boldsymbol{m}$ of solutions now includes the values for the previously filled cells and the value $\Delta\rho_j$ that is being tested:

$$m = (m_l)^T \text{ with: } m_l = \Delta\rho_j,\ m_l = \Delta\rho_l^+ \text{ if } l \in J_k^+,$$
$$m_l = \Delta\rho_l^- \text{ if } l \in J_k^-, \quad \text{otherwise } m_l = 0$$

.

To describe the solution, now we introduce the n-vectors $\boldsymbol{r}$, $\boldsymbol{g}$, $\boldsymbol{u}$, $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z}$ defined by the following ith components:

$$(\boldsymbol{x})_i = x_i - x_M,\ (\boldsymbol{y})_i = y_i - y_M,\ (\boldsymbol{z})_i = C_i, \quad (\boldsymbol{u})_i = 1,$$
$$(\boldsymbol{r})_i = \Delta g_i^c + A_{ij}\,\Delta\rho_j,\ (\boldsymbol{g})_i = \Delta g_i$$

and then we introduce the following notation

$$s_{mm} = \mathbf{m}^T\mathbf{Q}_{\mathrm{M}}^{-1}\mathbf{m} \quad \text{and} \quad s_{ab} = \mathbf{a}^T\mathbf{Q}_D^{-1}\mathbf{b}, \tag{10}$$

where the subscripts a and b and vectors $\boldsymbol{a}$, $\boldsymbol{b}$ correspond to pairs of subscripts r, g, u, x, y and z referring to the former vectors $\boldsymbol{r}$, $\boldsymbol{g}$, $\boldsymbol{u}$, $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z}$. With this notation, the normal equations corresponding to the system eqs (8) and (9) can be written as:

$$f s_{rr} + p_0 s_{ru} + p_x s_{rx} + p_y s_{ry} + p_z s_{rz} - s_{rg} + \lambda f s_{mm} = 0$$

$$p_0 s_{uu} + f s_{ru} + p_x s_{xu} + p_y s_{yu} + p_z s_{zu} - s_{gu} = 0$$

$$p_0 s_{ux} + f s_{rx} + p_x s_{xx} + p_y s_{yx} + p_z s_{zx} - s_{gx} = 0 \tag{11}$$

$$p_0 s_{uy} + f s_{ry} + p_x s_{xy} + p_y s_{yy} + p_z s_{zy} - s_{gy} = 0$$

$$p_0 s_{uz} + f s_{rz} + p_x s_{xz} + p_y s_{yz} + p_z s_{zz} - s_{gz} = 0.$$

The solutions of the equation system (11) can be calculated for the adopted prism and $\Delta\rho_j$ as:

$$f = (s_{rg} - p_0\, s_{ru} - p_x\, s_{rx} - p_y\, s_{ry} - p_z s_{rz})/(s_{rr} + \lambda\, s_{mm}),$$

$$p_0 = (F_{ug} - p_x\, F_{ux} - p_y\, F_{uy} - p_z F_{uz})/\, F_{uu},$$

$$p_x = (G_{xg} - p_y\, G_{xy} - p_z G_{xz})/\, G_{xx}, \tag{12}$$

$$p_y = (M_{yg} - p_z M_{yz})/\, M_{yy},$$

$$p_z = N_{zg}\,/\,N_{zz},$$

where the following abbreviation coefficients

$$F_{ab} = s_{ab}(s_{rr} + \lambda s_{mm}) - s_{ra} s_{rb}$$

$$G_{ab} = F_{ab}\, F_{uu} - F_{ua}\, F_{ub} \tag{13}$$

$$M_{ab} = G_{ab} G_{xx} - G_{xa} G_{xb}$$

$$N_{ab} = M_{ab} M_{yy} - M_{ya} M_{yb}$$

have been introduced.

Once the former linear equations have been solved, we can calculate the corresponding residual values v_i in terms of the selected $\Delta\rho_j$. Then, we take the global misfit value e_j^2 defined by:

$$e_j^2 = \mathbf{v}^T\mathbf{Q}_{\mathrm{D}}^{-1}\mathbf{v} + \lambda\, f^2\, \mathbf{m}^T\mathbf{Q}_{\mathrm{M}}^{-1}\mathbf{m} \tag{14}$$

as the parameter for the suitability of jth prism and the density possibility (positive $\Delta\rho_j^+$ or negative $\Delta\rho_j^-$) adopted. So, in this $(k+1)$th step, the method tests each of the unchanged prisms and both (negative and positive) density contrast possibilities. Then, the jth prism and selected density (positive and negative) which produce a minimum value of e_j^2, which we will call $E(k+1)$, are definitively selected to increase the model, adding their effect to the model values $\Delta g_i{}^c$.

This process is successively repeated. For the successive steps, the scale value f decreases and the additional parameters p_0, p_x, p_y and δ_{T} reach nearly stable values. The process stops when f approaches 1, resulting in the modelled structure for anomalous density, a final regional trend and a value (or a corrective one) for the terrain density.

Finally, the solution appears as a 3-D distribution of prismatic cells that have been assigned some of the prescribed contrast densities. Moreover, a regional trend and a terrain density are also obtained that supplements this anomalous mass distribution. Following this broad outline of the method, we now present now several comments and additional developments.

5 ADDITIONAL COMMENTS ON THE INVERSION PROCESS

To simplify the foregoing description of the inversion method, we have moved certain comments and additional explanations to this section.

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5.1 *A priori* anomalous density contrasts: smoothing of the 3-D model boundaries

The present inversion method requires, as usual for many non-linear methods, a suitable choice of the *a priori* density contrasts (positive and negative, in our case). An excessively high density contrast will fit the main anomaly component but will produce a rather 'skeleton' model, perhaps unable to represent small anomaly components. An excessively low density contrast will fit light structures, but would produce a rather 'inflated' model, perhaps unable (taking into account the volume limitations) to fit the main components, giving rise to fictitious peripheral bodies. To solve this issue, particularly in the case when knowledge of the true local density contrasts is not available, we adopt an alternative theoretical approach. Instead of invariable local density contrasts along the whole model formation process, we adopt contrasts that can vary. Starting from fixed maximum values (let us name R_j^+ positive and R_j^- negative for each jth cell), the prescribed density contrasts evolve along the growth process according to a simple law. By means of empirical tests, we have selected for each growth step the density contrasts, $\Delta\rho_j^+$ and $\Delta\rho_j^-$, positive and negative, given by:

$$\Delta\rho_j^+ = R_j^+ e^{-\tau f}, \quad \Delta\rho_j^- = R_j^- e^{-\tau f}, \tag{15}$$

where the scale factor, $f \geq 1$, corresponding to the step of the fit process is used as characteristic parameter to describe the growth instant and τ is a fixed factor corresponding to the desired variability of the density values. A high τ value will produce an anomalous model with contrast values mostly close to the maximum values (and then homogeneous with a sharp geometry). A low τ value will produce a model with more variable contrasts, decreasing outwards (and then with a more diluted geometry). In this form, the high-density contrasts (close to the maximum values R_j^+ and R_j^-) will be employed to suitably fit the main anomaly components, while smaller density contrast suitably fit the smaller and local anomalies. The resulting anomalous model is more versatile (and perhaps somewhat more diluted).

Let us observe that the process indicated here pursues an inverse objective to the one sought by the so-called 'flatness' constraint used in the linear problem inversion. Indeed, when the densities of the geometric elements (linear direct problem) are used as unknowns, the solution is too diluted or smooth, barely permitting specific bodies to be defined. Therefore, it becomes necessary to include a flatness condition based on the first derivatives of the density distribution, and that tends to favour better defined models (more homogeneous on the inside and with sharper density changes on the outside) (see Boulanger & Chouteau 2001; Bertete-Aguirre *et al.* 2002). In contrast, the arbitrary smoothness condition imposed here optionally permits the opposite effect: to slightly dilute the edges of the bodies.

5.2 Optimum balance between observed data fit and complexity of the model

The parameter λ governs the method minimization conditions regarding the balance between the model complexity (roughness, total anomalous mass) and observation data fit (mean residual). For small values of λ (and supposing a fine enough partition of the subsoil volume) a very complete data fit is obtained, and even part of the observational noise is fitted. However, in this case the anomalous mass model will be too massive and complex (there are fictitious masses, excessive peripheral masses, mixtures of positive and negative masses, etc.) because we are trying to invert part of the observational noise. Contrarily, for high λ values, the anomalous mass model is too condensed or simplified. In this case, the model cannot fit all the observed data, and the final residuals contain significant components of the gravimetric anomalies that have yet to be interpreted. Therefore, it is interesting to establish a criterion for choosing a λ_0 of balance between the data fit level and level of complexity of the model.

We suppose that the details of observation include standard deviation observational noise σ (in addition to some isolated outlier values that the method is able to detect and isolate). In these circumstances, the inversion process should invert 100 per cent of the data signal component, not 100 per cent of the data contents, leaving the amount corresponding to the observational noise as the final non-inverted noise. The right model should be complex enough to invert the signal component sign, but would not include fictitious masses in an attempt to justify noise components. So the fit level versus complexity level balance problem can be solved by analysing the possible inversion residuals in order to detect if they still include signal components suitable for inversion, or else it is merely uncorrelated noise.

To identify the signal-to-noise ratio in the final distribution of the inversion residuals, we use the covariance analysis technique (Moritz 1980). We suppose that the essential characteristic of the noise is the lack of spatial correlation between the values, while the correlated signals would constitute significant components worthy of inversion. Let v_i, $i = 1, \ldots, n$, be the final residuals in the respective points P_i. The empirical covariances as a function of the horizontal distance between the points will be given by:

$$\mathrm{cov}(d) = \frac{\sum_{\mathrm{i,j}} \tilde{v}_i \tilde{v}_j}{\sum_{k=1}^{\mathrm{n}} \tilde{v}_k \tilde{v}_k}, \tag{16}$$

where the $\tilde{v}_i$ are normalized residuals (in accordance with covariance matrix $\boldsymbol{Q_D}$ *a priori*), the sum of the numerator is extended to all the pairs of points P_i, P_j such that their mutual distance is close to d [dist .$(P_i, P_j) \approx d$], and the variance term of the denominator of (16) is used for standardization purposes.

Empirical covariance values can be determined for equally spaced horizontal distance values, $d_k = k\Delta d$, $k = 1, 2, \ldots$, where Δd is a suitable distance step. In particular, we take the median of the distances between each point and the three nearest points as the basic step Δd, that is, $\Delta d = \mathrm{median}\left\{\mathrm{dist}(P_i P_j);\ i = 1, \ldots n;\ P_j \text{ nearest three points to } P_i\right\}$. The empirical values $\mathrm{cov}(d_k)$, $k = 1.2, \ldots$, can in turn be fitted through an analytical covariance function (positive defined) (Barzaghi & Sanso 1983), for example, of the type:

$$C(d) = a\, J_0(cd)\, e^{-db}, \tag{17}$$

where $J_0(.)$ is the zero-order Bessel function and a, b and c are parameters to be determined.

Fig. 5 displays a covariance analysis for the residuals of the Faial inversion. Part (a) corresponds to the final residuals of an insufficiently developed model in which a correlated residual signal was observed. Case (b) corresponds to the optimum case of uncorrelated residuals. Case (c) corresponds to the final residuals of an overinverted model.

During the analysis, as the identifying characteristic parameter we choose the covariance at null distance $C(0)$, or else, for enhanced computational simplicity, its approximation by means of the first empirical covariance, $\mathrm{cov}(d_1)$. Therefore, the λ_0 value for the optimal model will correspond to a null value of $C(0) \equiv \mathrm{cov}(d_1)$.

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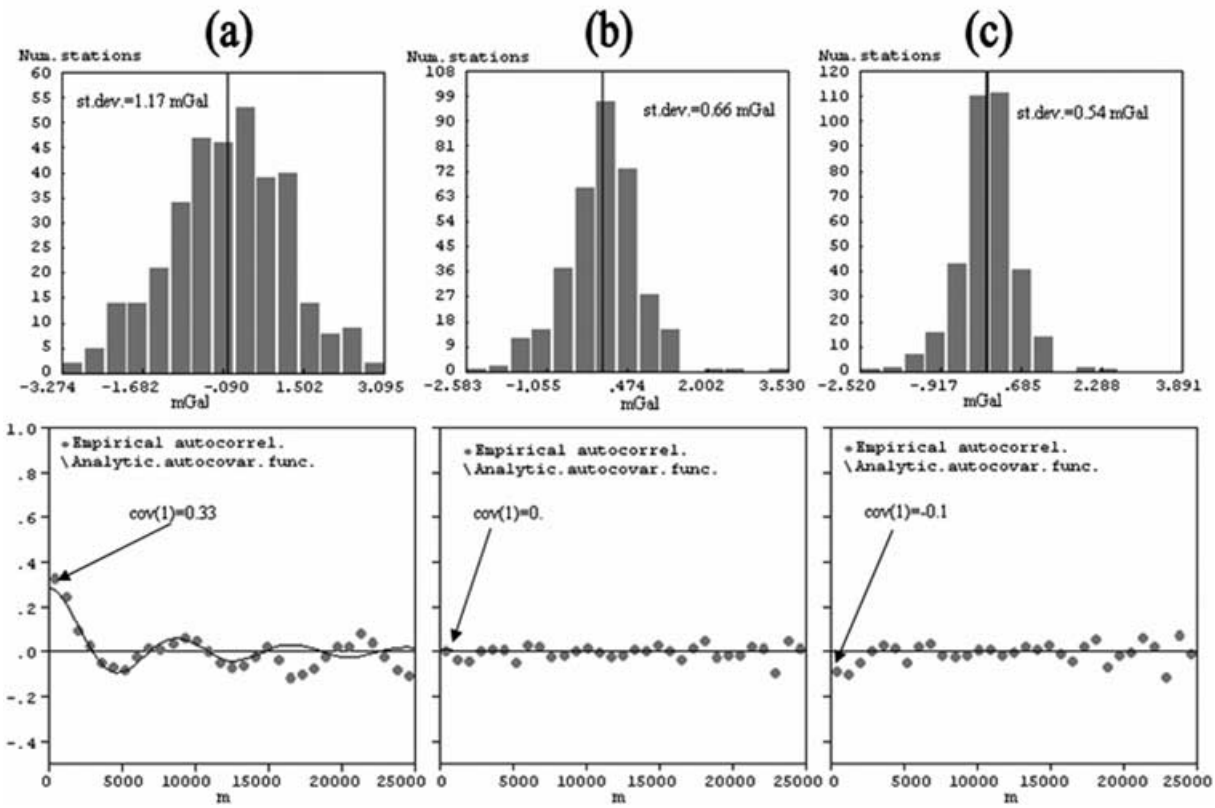

Figure 5. Histogram (up) and autocorrelation analysis (down) for the final residuals of the inversion of the gravity anomaly data from Faial. The empirical autocorrelation values were obtained with a step of 805 m. An analytic autocovariance function was also fitted. (a) Values for an oversimplified model, (b) values for an optimal model and (c) values for an overinverted model. The optimal case (b) is identified as producing a null first autocorrelation value: cov(1) = 0.

5.3 Treatment of errors in the gravity data

We must avoid possible data outliers in our inversion process. If all the residuals v_i arose from entirely random errors in the gravity data, they would have a normal distribution. In practice, some errors are due to instrumental defects or observer carelessness, and these are not normally distributed. In order to avoid their effect in the inversion model, outliers are assumed to be rare, so that the part of the combined distribution within $\pm B\sigma$, with $B = 2.5$, should be dominated by random errors. We assume a flat distribution for outliers (all sizes of outliers have equal probability), so this dominates in the region $>\pm B\sigma$ where the normal distribution is small. Then outliers could be detected by analysing the final residuals (Rousseeuw & Leroy 1987: Beltrao *et al.* 1991). Nevertheless, the step by step fit process lets us apply a weighting system designed to suppress the contribution by outliers to the adjusted model. First, in each step of the growth process, we use the median of the absolute values of the residuals as a robust estimation of the standard deviation:

$$\hat{\sigma} = \frac{\underset{i}{med}\,\{|v_i|\}}{0.6745}, \tag{18}$$

where the constant 0.6745 makes $\hat{\sigma}$ a consistent estimator of the standard deviation in the case of observations contaminated by Gaussian noise. With this deviation estimate, the standardized residuals $(v_i/\hat{\sigma})$ are computed and used to determine a suitable weighting system w_{bi} for observation g_i . A possible solution would be to set the w_{bi} equal to 0 in the region outside $\pm 2.5\hat{\sigma}$ and this would eliminate the outliers completely. One disadvantage of this approach is that during subsequent steps of the growth process, a gravity observation labelled as an outlier in the first steps, and then with zero weight, cannot be 'brought back into play' with full weight in the subsequent steps of the fitting process. A more flexible condition is to use a smoother filter, so that the weight is reduced from the value 1 if $(v_i/2.5\ \hat{\sigma}\ f^2)^2$ is greater than 1. The use of a quartic decay as:

$$w_{bi} = \frac{1}{1 + \left(\frac{v_i^2}{B^2\hat{\sigma}^2 f^2} - 1\right)^2} \quad \text{for} \quad v_i^2 > B^2\hat{\sigma}^2 f^2 \tag{19}$$

$$w_{bi} = 1 \quad \text{for} \quad v_i^2 \le B^2\hat{\sigma}^2 f^2$$

gives a smoothly tapering function for the outlier weight, which makes the convergence of the solution less erratic than with a step function. The w_{bi} is plotted in Fig. 6 for $f = 1$ as function of the standardized residuals $v_i/\hat{\sigma}$

The scale factor f has been included as an additional factor into the outlier weight system (19) to avoid premature false outliers. In fact, for the initial steps of the growth process, and taking into account the very simple initial structure, some large residuals can arise, which are then satisfactorily fitted by the model. This outlier control is performed through suitable bounding in the outliers detection. Then, we have selected bounding values given by $\pm c\hat{\sigma}$, where factor $c = f\,B$ takes into account the growth step by means of the model scale factor $f \ge 1$ and a general bounding value as $B = 2.5$. Now, Fig. 7 shows the evolution of the number of detected outliers across the model growth for the gravity data of Faial. Case (a) corresponds to a fit with an outlier weight system which does not

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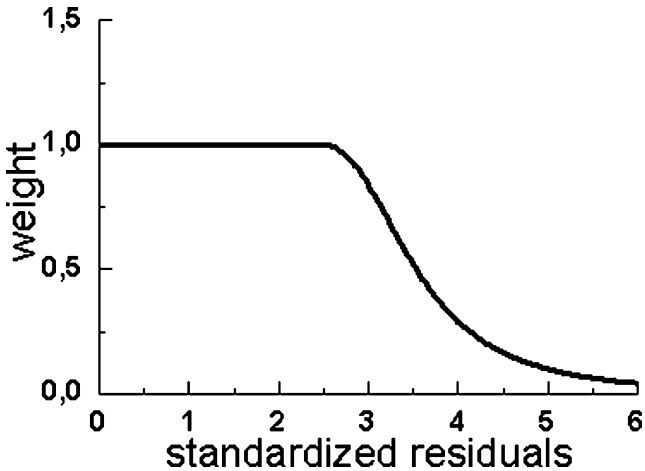

Figure 6. Weighting function adopted to reduce the effect of blunders corresponding to high values of the standardized residuals $v_i/\hat{\sigma}$. (a unit scale factor is considered).

include factor f, and case (b) corresponds to the outlier function given in eq. (19) including the scale factor. The number of outliers is mostly an increasing function, but without factor f (case a) a more erratic behaviour, with excessive and premature outliers, is observed.

If we consider an additional weighting w_s from an *a priori* estimated quality of the gravity stations, we can produce an expression for the combined weight $w_i = w_s w_{bi}$ for the case of *a priori* weights and suspected outliers.

5.4 Choice of the density for the terrain corrections

A traditional tool for estimating a suitable mean bulk density for terrain reductions was given by Nettleton (1939) from the gravity data themselves. The criterion was to seek a minimum correlation between gravity anomaly and topography, namely an independence between deep mass structures and topography. Nevertheless, for volcanic areas some correlation structure versus topography may be expected and, as a matter of fact, the ancient areas of the volcanic islands are located upon some high-density intrusive structures. On the other hand, sharp cones are constructed with light materials. Isostasy phenomena also correspond to some correlation between large relief and internal structure. In these circumstances, Nettleton's method does not give optimal results. A more suitable criterion can be deduced from the gravity inversion treatment.

Fictitious gravity anomalies are likely to occur if an inappropriate terrain density is used (for instance, too high a density gives rise to gravity lows over the mountains). Awkward fictitious bodies then result from inversion of gravity data with an inappropriate terrain density. In such circumstances, an interesting way of choosing the terrain's mean density could be to take a value that permits a good gravity fit while employing a small anomalous mass. These modelling conditions essentially match the conditions that determine our gravity inversion approach. Therefore, the inversion fit could include an additional parameter to prevent this uncertainty.

The computation normally used for the reduction of gravimetric observations due to the effects of the topographical masses is of the type $\rho_T\ C$ where ρ_T is the terrain mass density and

$$C = \pi G h + \sum_i C_i, \tag{20}$$

where G is the gravitation constant, h is the station altitude, and where the sum extends to the attractions C_i, for density one, of the geometric elements (triangular prisms, circular crown sectors, etc.) into which the volume of the topographical masses above and below the level of the station have been divided.

This computation can be performed *a priori* for C_i data from a digital model of the terrain and a known general value of ρ_T (for instance $\rho_T = 2670\ \mathrm{kg\ m^{-3}}$). However, due to the possible uncertainty of the value ρ_T as a local suitable value, it is interesting to include the computation of a corrective value $\delta\rho_T$ for mass correction at the same time as the inversion process. This gravimetric determination of $\delta\rho_T$ corresponds to the hypotheses put forward at the start of this paragraph, namely best-fitting and simplest anomalous mass model. It is performed by including the mass correction term δg_{top} in the gravimetric anomalies expression, and maintaining the value of $\delta\rho_T$ (corrective from a provisional value) as the unknown value, which will be determined in the inversion process. The approach is similar to the one applied to the regional trend term. Nonetheless, it should be noted that this gravimetric type of determination works well when the topographic effects are evident (if the terrain is flat, we must remove the term $\delta g_{\mathrm{top}} = \delta\rho_T C$ in eq. 2) and also if we assume the general minimization conditions of the inversion process.

5.5 Regional trend

The regional trend present in gravimetric data is usually fitted by means of a specific preliminary study. A usual procedure is polynomial adjustment (see Beltrao *et al.* 1991, for a robust polynomial adjustment). Here we propose a method to determine the regional trend that is integrated and simultaneous with the inversion process.

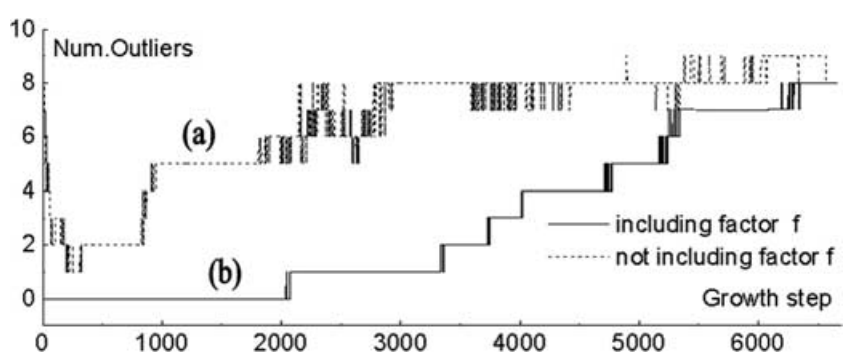

Figure 7. Evolution of the number of outliers across the model growth (data from Faial). (a) Dash line, without scale factor, (b) solid line, including the scale factor $f \geq 1$ into the weighting formula.

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Perhaps the main problem lies in the exact definition of the regional trend. In this simultaneous adjustment (for inversion model, regional trend, terrain density and observational noise), the regional component of the anomalies is conceived as that part of the data which, due mainly to its long wavelength, cannot be modelled with the volume elements predicted by the model. Therefore, the regional condition is of the 'negative' type: non-local, non-adjustable to the local structures in play. This condition can only be met in a simultaneous fit. (Under these circumstances, and as we will see in the simulation section, there may be a certain exchange between the regional component and the anomalies due to the model's deeper structures.)

Here we have used a linear representation $\delta g_{\mathrm{reg}} = p_0 + p_x(x_i - x_M) + p_y(y_i - y_M)$, $i = 1, \ldots, n$, of the regional component, to simplify the method's formulism. p_0 is a constant base and p_x, p_y are SN and WE horizontal gradients. With a little more hard work, it is possible to develop, for example, the second degree formula.

5.6 Estimation of the precision for the adjusted parameters

As useful information, we can calculate values for the estimated precision of the parameters resulting for the inversion approach (anomalous density distribution, regional trend and correction for the terrain density). For that, we write eqs (2) and (8) as:

$$\mathbf{d} = \mathbf{G}\,\mathbf{p} - \mathbf{v}, \tag{21}$$

where $\boldsymbol{d} = (\Delta g_1 \ldots \Delta g_n)^{\mathrm{T}}$ is the data vector, $\mathbf{p} = (\Delta\rho_{1_1}, \ldots, \Delta\rho_m, p_0, p_x, p_y, \delta\rho_{\mathrm{T}})^{\mathrm{T}}$ is composed by the model parameters ($\boldsymbol{m}$ and the trend and terrain parameters) and $\boldsymbol{G}$ is the $n \times m$ matrix for these linear equations (with $n < m$) as extension of matrix $\boldsymbol{A}$. The covariance matrix for the data vector $\boldsymbol{d}$ is again $\boldsymbol{Q_D}$, but now the covariance matrix $\boldsymbol{Q_P}$ for the parameters $\boldsymbol{p}$ can be obtained from matrix $\lambda\boldsymbol{Q_M}$ by adding the *a priori* variances for p_0, p_x, p_y and $\varepsilon\rho_{\mathrm{T}}$. Then, the posterior covariance matrix $\boldsymbol{Q}_{\mathbf{P}}$ for parameters $\boldsymbol{p}$ can be obtained as (Tarantola 1988):

$$\mathbf{Q}_{\mathbf{P}'} = \mathbf{Q}_{\mathbf{P}} - \mathbf{Q}_{\mathbf{P}}\mathbf{G}^{\mathbf{T}}\left(\mathbf{G}\mathbf{Q}_{\mathbf{P}}\mathbf{G}^{\mathbf{T}} + \mathbf{Q}_{\mathbf{D}}\right)^{-1}\mathbf{G}\mathbf{Q}_{\mathbf{P}} \tag{22}$$

6 SIMULATION TEST

Here we will consider a simulation by means of an arbitrary example that lets us highlight the methods' advantages and constraints. To better approximate the actual data under study, we will suppose the same topographical relief and the same distribution of observation stations. However, we will suppose an arbitrary structure of anomalous masses that includes both positive bodies (excess mass, with contrast 500 kg m^{-3}) and negative bodies (defect of mass, with contrast -600 kg m^{-3}). This structure, displayed in Fig. 8, has a total anomalous mass of 85.5×10^{12} kg (45.0×10^{12} kg for the negative masses and 40.5×10^{12} kg for the positive masses) and its centre of masses is located at an average depth of -3605 m (-3300 m for the negative masses and -3944 for the positive masses). To the gravity attraction values of these bodies, and to form the 'observation data', we add (1) an arbitrary trend $p_0 = 100.0 \times 10^{-5}$ m s^{-2}, $p_x = 0.40 \times 10^{-8}$ m s^{-2} m^{-1}, $p_y = 0.0$ m s^{-2} m^{-1} and (2) a topographical terrain density effect with $\rho_{\mathrm{T}} = 0$ kg m^{-3}. Fig. 9 shows the simulated anomaly.

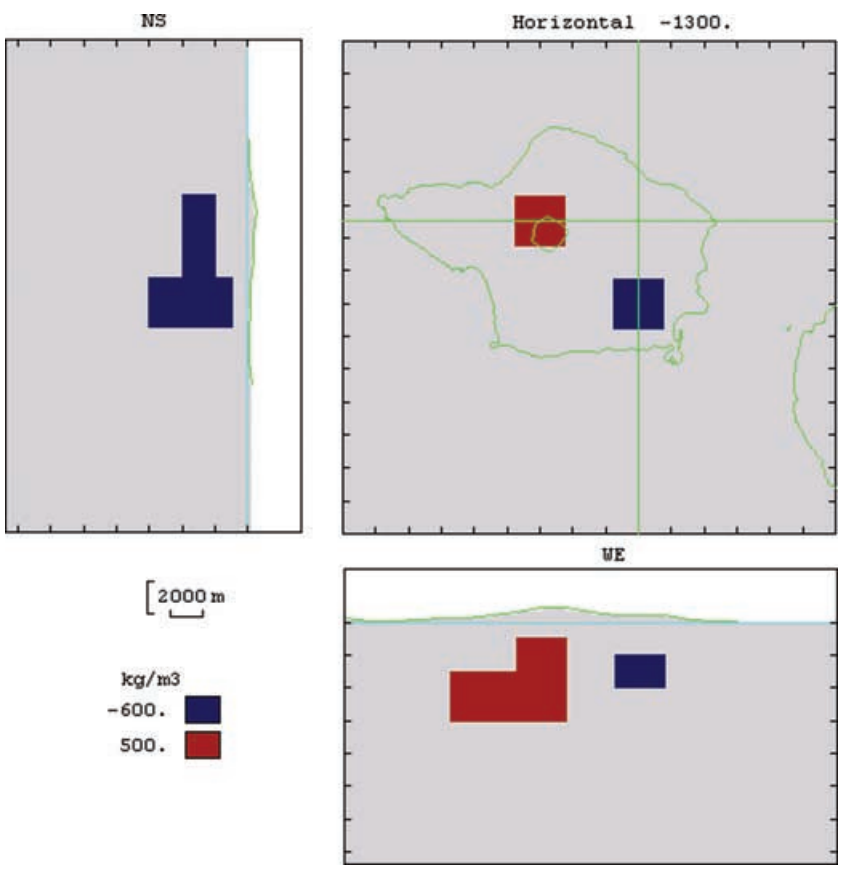

Figure 8. Simulation model constituted by two arbitrary bodies with -600 kg m^{-3} (blue) 500 kg m^{-3} (red) buried below the same stations in Faial. A simulated additional trend is also considered.

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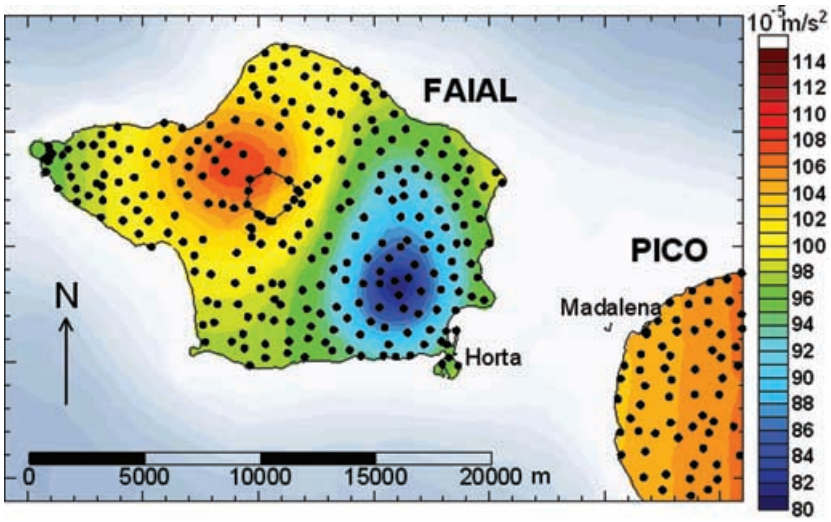

Figure 9. Bouguer anomaly corresponding to the simulation model of Fig. 8 and a simulated trend $p_0 = 100.0 \times 10^{-5}$ m s^{-2}, $p_x = 0.40 \times 10^{-8}$ m s^{-2} m^{-1}, $p_y = 0.0$ m s^{-2} m^{-1} for the same stations of Faial. Tick marks in axes correspond to 1 km.

First of all, we carefully partition the local subsurface volume into prismatic cells with sides of the order of 300 m. (From an operational perspective, it is better to use an initial partition of larger cells and, once a first model has been obtained, to use a more local partition with small cells.) Secondly, we choose a few *a priori* density contrasts, for example $R_j^+ = 700$ kg m^{-3} and $R_j^- = -460$ kg m^{-3}, and an average factor $\tau = 5$ for smoothing the mass transition on the edges of the bodies. Having established this, the procedure is automatic.

Fig. 10 displays profiles of the physiognomy of the fitted inversion model. The fitted values for the different parameters are as follows: $p_0 = 100.19 \times 10^{-5}$ m s^{-2}, $p_x = 0.398 \times 10^{-8}$ m s^{-2} m^{-1}, $p_y =$

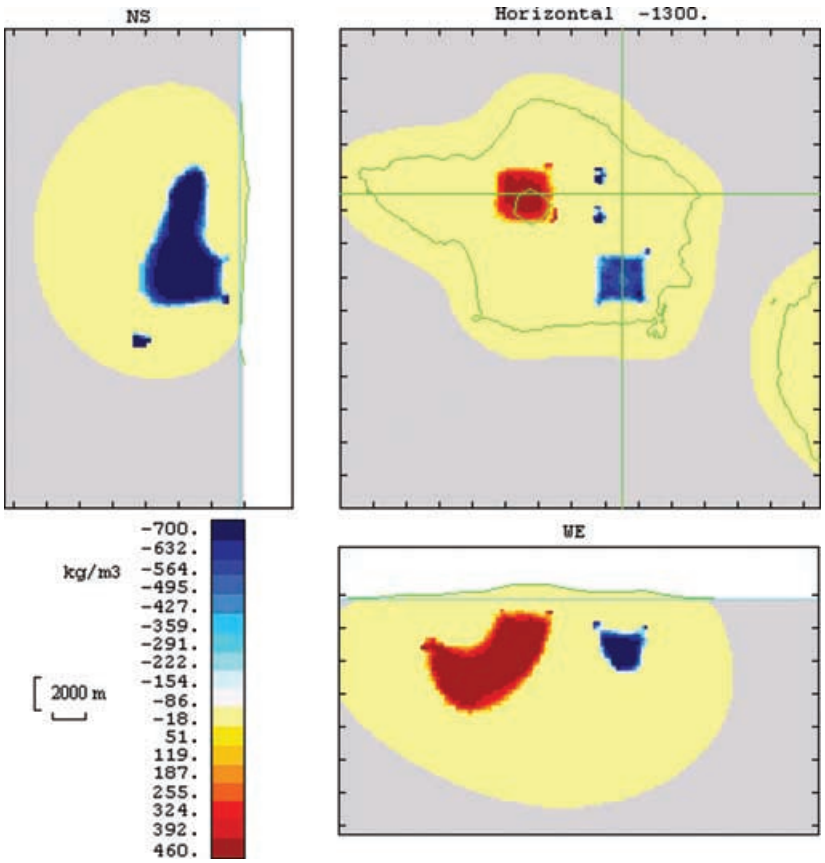

Figure 10 3-D model of anomalous density obtained by means of inversion of the simulated data of Fig. 9 and by assuming extreme density contrasts of −700 and 460 kg m^{-3}. Some rounding effect is observed for the volume in the bottom of the model. The adjusted values for the trend were 100.19×10^{-5} m s^{-2}, $p_x = 0.398 \times 10^{-8}$ m s^{-2} m^{-1}, $p_y = 0.001 \times 10^{-8}$ m s^{-2} m^{-1}.

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0.001×10^{-8} m s^{-2} m^{-1}, $\rho_T = 1$ kg m^{-3}, $\sigma = 0.003 \times 10^{-5}$ m s^{-2}, total anomalous mass $= 81.7 \times 10^{12}$ kg, negative $= 43.7 \times 10^{12}$ kg, positive $= 38.0 \times 10^{12}$ kg, total average depth $= -3510$ m, negative masses $= -3257$ m and positive masses $= -3801$ m. We see that, for this case of no errors, results are quite good.

Now we simulate an observational noise by adding a Gaussian noise of arbitrary standard deviation 0.300×10^{-5} m s^{-2} (5 per cent from the data variance) to the previous simulated data. Moreover, we include in the data three arbitrary outlier values. This additional distortion produces a distortion and a reduction of the inverse model. In fact, the basic covariance sampling distance is $\Delta d = 805$ m, which, as we indicated earlier, is the mean of the distances between neighbouring points. It provides us the estimation of a balance parameter for the inversion. By applying the inversion process, now we obtain: $p_0 = 100.88 \times 10^{-5}$ m s^{-2}, $p_x = 0.362 \times 10^{-8}$ m s^{-2} m^{-1}, $p_y = 0.043 \times 10^{-8}$ m s^{-2} m^{-1}, $\rho_T = 42$ kg m^{-3}, not so good values as for no observational errors. The resulting standard deviation of the final residual is $\sigma = 0.3047 \times 10^{-5}$ m s^{-2}, quite similar to the simulated one. Moreover, the three outlier data are clearly detected and discarded. The resulting 3-D model is significantly reduced (especially in the deeper and less quality zones): total anomalous mass $= 66.1 \times 10^{12}$ kg, negative $= 44.0 \times 10^{12}$ kg, positive $= 28.1 \times 10^{12}$ kg, total average depth $= -3122$ m, negative masses $= -3148$ m and positive masses $= -3072$ m.

The following comments may be made in view of these results.

1. The method works satisfactorily in the context of a fit with very free conditions (no additional information or constraints) and large number of parameters.
2. As is generally predictable for potential fields, with uncertain surface gravity data, it is almost impossible to fully reproduce the deep structure (except with very exhaustive additional information), and especially in cases such as this that consider models that include both excesses and defects of mass.
3. Also predictably, it is observed that the model tends to 'smooth' the corners. This smoothing effect is evidently greater in the deeper areas than in the shallow ones.
4. The presence of observational noise produces some distortion and reduction of the model, mostly in the deeper and bad determined zones.

7 INVERSION MODEL FOR FAIAL AND DISCUSSION

The input gravity data is described in Section 3. There are not enough precise models about the inner structure of the island in the published literature to be used as initial model for our inversion approach. Then we carried out a free adjustment corresponding to the observed gravity data. To that end, we considered a subsurface volume partition with cells of side ranging from 106 m for shallowest cells to 665 m for the deepest cells (looking for a similar mean square effect on the observation points). The maximum density contrast allowed was -400 and 300 kg m^{-3} everywhere, and the smoothness coefficient was $\tau = 5$ (see eq. 14).

After testing several tentative inversion models, we obtained (see Fig. 5) a suitable parameter λ_0 for optimal balance between model fitness and model magnitude according the proposed methodology. The inversion process simultaneously determined: a linear regional trend for the gravity data described by a decreasing rate of 0.59×10^{-8} m s^{-2} m^{-1} towards the southwest extreme (N224°E), a terrain density of 2341 kg m^{-3}, a data noise level of 0.59×10^{-5} m s^{-2}, and an anomalous model with mean density contrasts of -347 and 268 kg m^{-3}. The estimated error values for these parameters were $\pm 147 \times 10^{-5}$ m s^{-2}, $\pm 22 \times 10^{-8}$ m s^{-2} m^{-1} and $\pm 42 \times 10^{-8}$ m s^{-2} m^{-1} for the components p_0, p_x and p_y of the trend, ± 21 kg m^{-3} for the terrain density and a mean value of ± 92 kg m^{-3} for the anomalous density of the cells. The regional trend agrees with the forecasted gravity trend in values that decrease towards the oceanic ridge. The terrain density value is close to the value of 2350 kg m^{-3} used by Demande *et al.* (1982) and lower than the value of 2430 kg m^{-3} given by Nettleton's method. The anomalous density model is formed by 17 239 cells (3296 of them are filled cells) that fill the subsurface volume up to a maximum depth of 6 km. Figs 11 and 12 show horizontal and vertical slices through the 3-D model.

Several main features can be observed in the inversion model:

(1) A large high-density body, which we name body *A*, is located under the eastern half of the island. It resembles a wall-like structure that starts near the central caldera and follows a N100°E azimuth towards the eastern shore. An anomalous mass of about 22.5×10^{12} kg is estimated for this body. For the adopted density contrasts, the dimensions of body A are approximately (Figs 11 and 12): depth of bottom 6 km below sea level, depth of top 0.5 km, length 9 km, mean width 2 km, volume 75 km^3 and depth of the mass centre 3.2 km. Despite the strong tendency of the inversion method to produce rounded bodies (especially for deeper areas), a wall-like structure is clearly identified in our model.

(2) A second high-density body, named body B, is located west of the central caldera. It also is elongated but has a SW trend (azimuth N220°E), until it reaches the shore near Varadouro. The estimated anomalous mass of body B is 4.8×10^{12} kg and it is at a shallower depth than body A, between 0.5 km a.s.l. and 3 km b.s.l., with depth of mass centre 1.5 km b.s.l.

(3) There are some other smaller, shallower positive bodies whose properties are close to the limit of the model resolution (due to the noise level and to the gaps between stations). For example, there is a small body, named *C*, with an elongated structure that starts at the western rim of the caldera (northwestern main structure of body *B*) and then follows a N300°E azimuth towards the northwest shore (Figs 11 and 12). The depth of its mass centre is about 0.8 km b.s.l., and together with body *B* defines a horseshoe-shaped structure, best seen in sections -500 and -800 m of the inversion model. The location and shape of the horseshoe-shaped high-density body are coincident with the boundary between the Capelo Volcanic Complex (of Holocene age) and the 'Caldeira' Volcano (see Fig. 3). Another smaller, very thin and elongated positive structure, named D, is located at very shallow depth in the graben area, following nearly parallel courses N119°E, which is also the trend of Pedro Miguel Graben main faults (Fig. 3).

(4) The low-density bodies are mainly located in peripheral positions, so their geometry cannot be clearly established from our terrestrial data. However, some features can be described. First, a very strong negative body ('*a*' on Fig. 12) is modelled at the western edge of the island and coincides with the Capelinhos Volcano, which erupted in 1957–1958 A.D. Further conclusions about this body's shape and dimensions are not possible at the present time. We would only highlight the shallow elongation of this body, inland of Faial Island, coincident with the trend of the volcano-tectonic lineament defined by the scoria cones of Capelo peninsula, which is close to N115°E.

(5) Other much smaller negative bodies are located in shallow peripheral positions, such as north of Ribeira Funda and at Monte da Guia sea shore (bodies '*b*' and '*c*' on Fig. 11).

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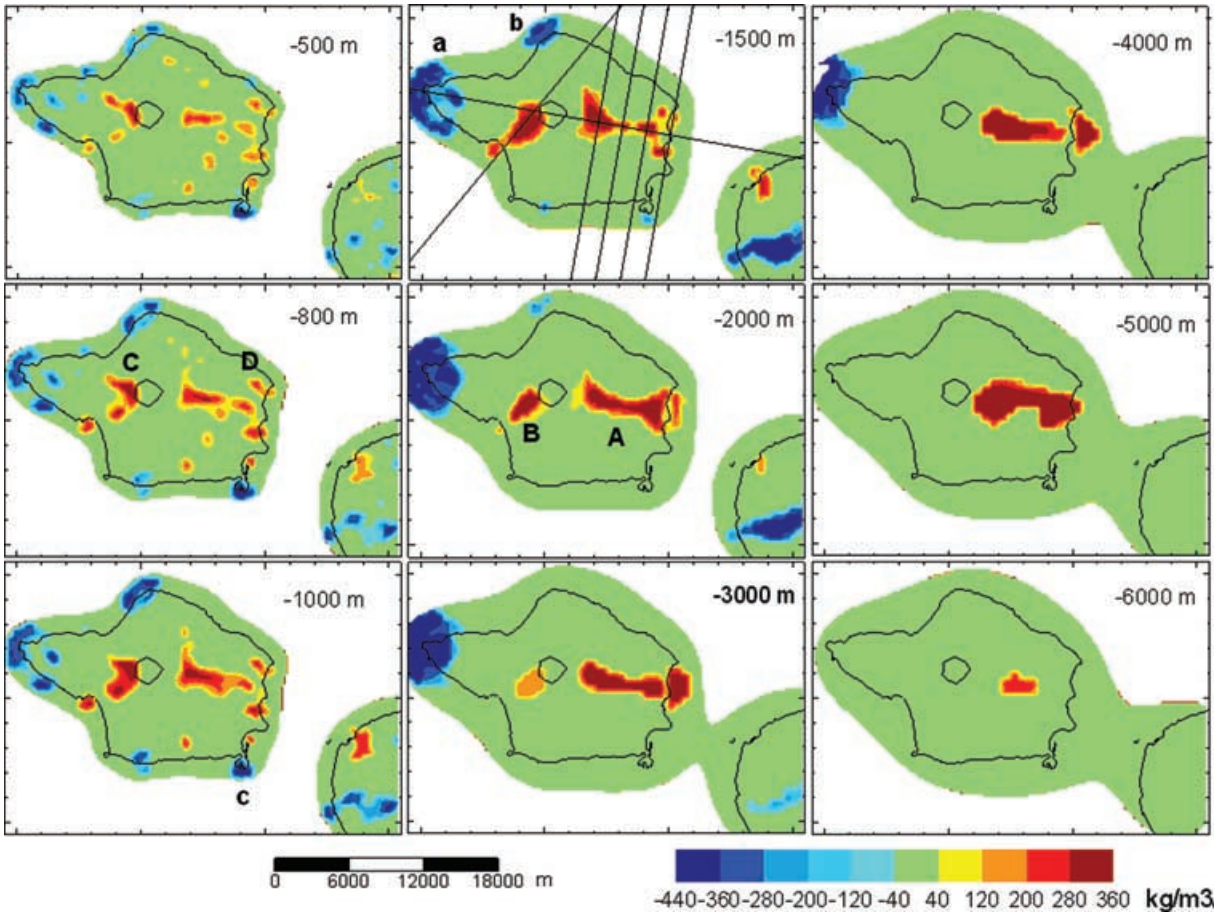

Figure 11. Horizontal sections of the anomalous model corresponding to depths ranging from −500 to −6000 m. Section −1500 m shows the location of the cross-sections presented in Fig. 12.

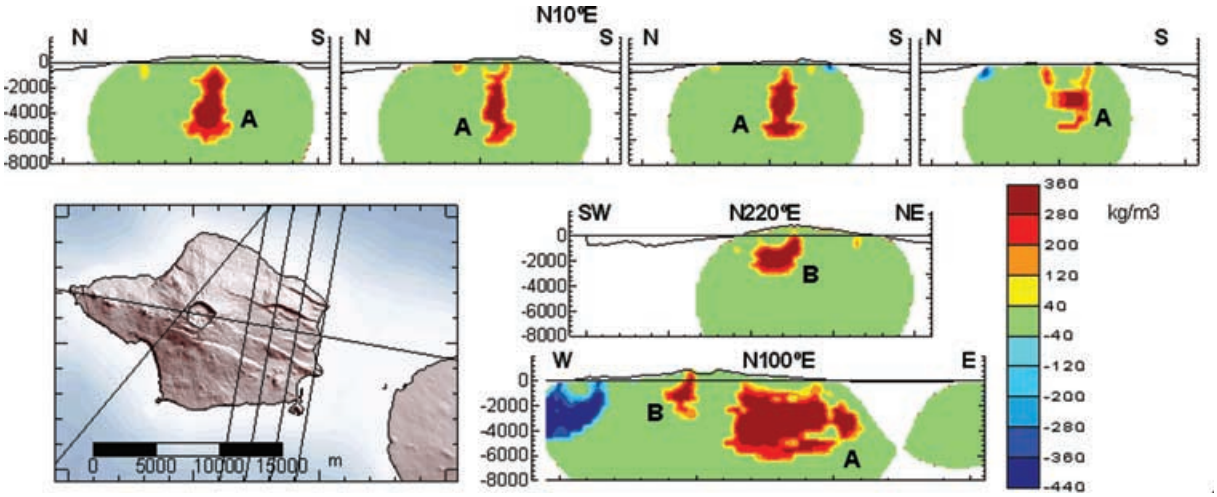

Figure 12. Vertical profiles N10°E, N220°E and N100°E and of the inversion model showing the main intrusive body A (red) with positive density contrast.

We interpret our results in light of the ideas of Walker (1999) and Rymer & Brown (1986), given in the introduction. The centre of a mature oceanic shield volcano is characterized by a column-like body of high-density rocks. These high-density rocks produce positive gravity anomalies because of the density contrast between the dense intrusive complex or magma body and the lighter, vesiculated and uncompacted eruptive materials. Due to the strong control of the regional tectonics (dominantly WNW–ESE) on the volcanism in Faial Island (Madeira 1998), an elongated form along the tectonic trend could be expected for such a high-density structure. Body *A* has the expected geometry and trend of a large intrusive body comprised of a compact swarm of hundreds/thousands of dykes and sheet-like intrusions. Body A may thus represent the main deep feeder system of the old Ribeirinha shield volcano located east of the caldera

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(Fig. 3), which is concealed by recent deposits, namely from 'Caldeira Volcano'. The base of this intrusive complex is located at a depth of 6 km, which is about half of the crustal thickness (10–12 km) in the area of the Azores Plateau (Luis *et al.* 1998).

Nunes *et al.* (2006) came to a similar conclusion for the Topo shield volcano area, in Pico Island. The Topo shield volcano is also characterized by a prominent gravity positive anomaly. The source of the anomaly extends to depths of about 8 km (mean depth of 4.5 km) and has a volume of 310 km^3. The body may reflect the major intrusive system associated with the Topo volcano. The body also coincides with gravitational collapses (e.g. subsidence phenomena) of the volcano surface, associated with a loading effect by the volcanic edifice due to its growth.

We interpret that body A is related to the early stages of the Faial island eruptive history and is the main deep feeder system of an old shield volcano that today crops out on the eastern sector of Faial island (the Ribeirinha Volcano, Fig. 3). The trend of the body appears to be controlled by the regional tectonics. Its clearly elongated shape suggests a narrow rift zone throughout a dilated effusive episode. New magma batches under the rift zone produced growth of the complex at the edges and lateral surfaces in line with neutral buoyancy levels. Dias & Matias (2006), using seismic methods, detect the presence of a similar structure located at mid-lower crust and coincident with the NNW–SSE orientation of the most active seismogenic structure. According the authors, this body may correspond to a plutonic intrusion that used a pre-existent deep fault to its emplacement. They accept also the possibility of a magmatic chamber beneath the main volcanic structures.

Furthermore, the cooled intrusive complex has suffered subsidence, helped by local volcanic and seismic activity. The well-defined graben in the eastern half of Faial Island (Pedro Miguel Graben) is the result of this subsidence process, which started about 73 000 years ago. Given that body *A* probably continues eastwards under the sea, to the Madalena area, on Pico island, it is also likely that the graben extends also eastwards to the neighbouring island of Pico, although covered by the young (Holocene) lava flows of Pico Mountain Volcano (Nunes *et al.* 2006).

The trend of body *A* (azimuth N100°E) does not match exactly the general trend of the Pedro Miguel Graben fault scarps and the dominant volcanotectonic lineaments of Faial island, with azimuth N115°E, suggesting some clockwise rotation of the active lineaments.

The central caldera is located at the western edge of the intrusive body *A*. Body *B* (and minor body *C*) appear to be intrusive bodies that are not directly connected to the main regional structure at Faial island, but may have propagated from the central volcano along subhorizontal buoyancy lines. It is likely that body *B* would correspond to an ancient large intrusive episode.

The adjusted inversion model has several restrictions and limits. First, as usual for inversion of potential fields, there is a non-uniqueness problem. The final solution must be considered as one possible good solution (that one involving an optimal balance between data fit and model roughness for the prescribed density contrasts). Second, the model depends on the suitability of the prescribed density contrast; some testing, however, can be applied to check possible density contrasts. Third, we know that a large noise level in the data can give rise to some distortion and size reduction in the model. Fourth, the quality of the model is not the same everywhere. The deeper and peripheral zones are poorly determined. The downward ('roots') and lateral continuation of the modelled structures is not revealed. Finally, the model, which in the figures appears to be composed by certain bodies floating in a homogeneous 'background' medium, must be taken as representing an anomalous structure with respect to any background structure. This background structure could be any mass distribution, such as a homogeneous medium or any horizontally stratified medium that does not produce horizontal gravity gradients. It could be provided by another geophysical technique sensitive to horizontal stratification, for instance, a seismic tomography.

8 CONCLUSIONS

From a methodological point of view, we give an expanded description of a general method for 3-D gravity inversion previously introduced in earlier papers. The aim of the method is to find a model of anomalous density contrasts (positive and negative contrasts) in an exploratory process based on a growth building of the bodies. Here we made several improvements to the inversion process. First, a robust treatment of data provides a tool for identifying and controlling possible outlier gravity values (mainly stemming from errors in gravimetry, positioning, altimetry or terrain corrections). Error detection is accomplished simultaneously with the main task of the gravity study. Second, we propose the introduction of a new degree of freedom for the inversion fit, that enables us to determine (or correct) a suitable density value for topographic mass reduction. This purely gravimetric estimation of density is incorporated in the global fit and based on the inversion conditions: small fit residuals and also small anomalous mass of the model (thus avoiding fictitious additional bodies corresponding to inadequate terrain density). Third, here we introduce a criterion to decide a suitable value λ_0 for the balance between model fitness and model magnitude (total anomalous mass). For that purpose, the data noise level is determined by means of a covariance analysis of residuals. The inversion process chooses the model that produces a zero covariance level in the residuals, avoiding fictitious bodies for inversion of noise components in the data.

We apply this inversion method to a gravity data set that covers Faial Island. We determine a mean terrain density of about 2341 kg m^{-3}, enabling us to calculate the Bouguer gravity anomaly and produce a map of the area with an estimated noise of 0.6×10^{-5} m s^{-2}. The other result produces a model of the anomalous density structure beneath the island.

The main feature of the inversion model is a high-density body that has a wall-like geometry. The body strikes N100°E from the centre of the island of Faial towards Madalena area, on Pico Island. The body, located at depths between 0.5 and 6 km b.s.l. below Faial Island, is interpreted to be a compact dyke swarm, formed along a regional tectonic structure, the Faial-Pico Fracture Zone. The intrusive complex was formed during the early stages of the Faial eruptive history (about 730 000 years ago), and is interpreted to be the main deep feeder system of the old Ribeirinha shield volcano on the eastern part of Faial island and exposed between Espalamaca and Ribeirinha. The gravimetric data also allowed us to locate the central area of that shield volcano, in the vicinity of Cabeço da Rocha Alta (the neck and remains of an old scoria cone), within the Pedro Miguel Graben.

The cooled intrusive complex has suffered a large amount of subsidence (of about 170 m) during the last 73 000 yr as manifested by a series of down-dropped fault blocks on the eastern part of Faial Island. It is not clear if the graben is a purely tectonic structure (as the result of stretching of the surface of the volcano in relation with the regional stress field), or the result of the removal of magma from underlying intracrustal magma reservoir(s). The general

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westward migration of the main volcanic centres on Faial island (from Ribeirinha shield volcano, to 'Caldeira' central volcano and then to 'Capelo' peninsula), suggests a westward propagation of the underlying main deep reservoirs and feeder system. If so, the concomitant withdrawal and instability on the volcano surface could be called to explain that subsidence in the eastern, older, part of Faial Island.

Other much smaller intrusive bodies defined by the inversion model were identified in shallow areas and were interpreted as structures that propagated laterally from the main volcanic centres and following neutral buoyancy levels. These bodies may correspond to later volcanic stages of Faial island eruptive history.

ACKNOWLEDGMENTS

This paper is a contribution to the project PPERCAS 'Estudo do Risco/Casualidade Sísmica do Grupo Central do Arquipélago dos Açores', contract PRAXIS/Ciência e Tecnologia, nr. 3/3.1/CEG/2531/95, to the project VULCMACII MAC/2.3/A7 of the European Union Programme INTERREGIIB and to the project 'Estudio gravimétrico conjunto de las Islas de Faial, S.Jorge y Pico. Estructura cortical obtenida a partir de técnicas de inversión. Contribución al conocimiento de la geodinámica en la zona del Punto Triple de Azores', contract MCYT/Acciones Integradas Hispano-Lusas, nr. HP2001-00810081 and contract CRUP/Acções Integradas Luso-Espanholas, nr. E 61/02. This work has been partially supported by research project REN2002-03450. We thank Hajo Götze and Carl Gerstenecke for their useful suggestions to improve the quality of this manuscript.

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